

Event-Driven Spacecraft Pose Estimation via CNN Direct Regression from Voxel Grids

Anonymous SPARKS@CVPR submission

Paper ID XXXX

Abstract

001 We present a deep learning pipeline for 6-DoF spacecraft
002 pose estimation from event-camera data, developed for the
003 SPARKS Challenge at CVPR 2026. Given a stream of asyn-
004 chronous events from a 1280×720 neuromorphic camera, we
005 convert each 100 ms event window into a compact 3-channel
006 exponential-decay voxel grid and regress pose directly us-
007 ing a ResNet-50 backbone with dual translation and rota-
008 tion heads. Rotation is supervised with geodesic distance
009 on $SO(3)$ rather than Euclidean quaternion loss. We com-
010 pare three architectural variants—a frame-by-frame CNN, a
011 CNN+GRU temporal model, and a heatmap-based keypoint
012 regressor with PnP—on the SPADES dataset (300 synthetic
013 training sequences, 31 real test sequences). Our best sub-
014 mission achieves a composite pose error of **2.063** (transla-
015 tion 0.4105, orientation 1.6526 rad) on the final leaderboard.
016 Analysis reveals that the synthetic-to-real domain gap, driven
017 primarily by a $100 \times$ timestamp-unit mismatch between train-
018 ing and test data, is the dominant performance bottleneck,
019 and that single-frame prediction outperforms temporal mod-
020 elling at the 10 Hz pose sampling rate used in this dataset.
021

022 1. Introduction

023 Accurate knowledge of a spacecraft’s pose—position and
024 orientation relative to a camera—is essential for autonomous
025 rendezvous, proximity operations, and on-orbit servicing.
026 Traditional frame-based cameras suffer from motion blur
027 and limited dynamic range in the high-contrast conditions of
028 space, motivating the use of *event cameras* [2]. Event cam-
029 eras report per-pixel brightness changes asynchronously with
030 microsecond temporal resolution, offering high dynamic
031 range, low latency, and low power consumption. These prop-
032 erties are well suited to the rapid rotational motions and
033 extreme illumination conditions encountered in orbit.

034 The SPARKS challenge [16] provides the SPADES
035 dataset: 300 photorealistic synthetic training sequences ren-

dered with event-camera simulation, paired with 31 real
event-camera recordings of a Proba-2 spacecraft mockup.
The task is to predict the 6-DoF pose (translation $\mathbf{t} \in \mathbb{R}^3$ and
unit quaternion $\mathbf{q} \in \mathbb{H}$) at 10 Hz from the raw event stream.

We develop an end-to-end CNN pipeline and explore
three directions:

- 042 1. **Frame-by-frame regression:** A ResNet-50 backbone
043 with dual regression heads trained on 3-channel event
044 voxel grids.
- 045 2. **Temporal modelling:** A CNN+GRU network that aggre-
046 gates context over 10 consecutive frames.
- 047 3. **Keypoint + PnP:** A heatmap decoder regressing 2D pro-
048 jections of known 3D spacecraft keypoints, followed by
049 iterative PnP.

Our main findings are: (i) frame-by-frame CNN prediction
outperforms temporal aggregation at 10 Hz; (ii) rotation error
is substantially harder to close than translation error across
the synthetic-to-real gap; and (iii) a $100 \times$ timestamp-unit
mismatch between training and test data is a key source of
domain shift that future work should address explicitly.

056 2. Related Work

057 **Event camera representations.** Raw events cannot be
058 directly ingested by conventional CNNs. Zhu *et al.* [18] in-
059 troduced the *voxel grid*, which distributes events into tempo-
060 ral bins via bilinear interpolation, preserving both temporal
061 structure and polarity. Alternative representations include
062 time surfaces [7], event frames [13], and learned event-to-
063 frame conversions [14]. We use a signed voxel grid and
064 a three-channel exponential-decay variant that is directly
065 compatible with ImageNet-pretrained RGB backbones.

066 **Spacecraft pose estimation.** Classical model-based track-
067 ers (e.g., IPPE, EPnP) require textured CAD models and
068 struggle under event-camera noise. Deep learning ap-
069 proaches for frame-based cameras include direct regression
070 [6] and keypoint-based methods [5, 9]. The SPEED [15]
071 and SPEED+ [10] datasets catalysed learning-based meth-
072 ods for spaceborne pose estimation, but use standard cam-

073 eras. SPADES [12] is, to our knowledge, the first large-scale
074 dataset pairing event cameras with 6-DoF spacecraft pose
075 labels.

076 **Loss functions for rotation.** Naive ℓ_2 loss on quaternion
077 components is non-geometric and breaks under the double-
078 cover of $SO(3)$. Geodesic distance, $2 \arccos(|\hat{\mathbf{q}} \cdot \mathbf{q}|)$, is the
079 natural metric on the unit sphere and has been shown to
080 improve convergence for rotation regression [8, 11]. We
081 adopt geodesic rotation loss throughout.

082 **Domain adaptation.** The DANN framework [3] uses a
083 gradient-reversal layer to learn domain-invariant features.
084 We incorporate DANN into our CNN+GRU model to en-
085 courage synthetic-to-real transfer.

086 3. Method

087 3.1. Event Representation

088 Each pose timestamp τ is associated with a window of events
089 $\mathcal{E}_\tau = \{(x_i, y_i, p_i, t_i)\}$ spanning $[\tau - 100 \text{ ms}, \tau]$, where
090 x_i, y_i are pixel coordinates, $p_i \in \{-1, +1\}$ is polarity, and
091 t_i is timestamp.

092 **3-channel exponential decay (our primary representa-**
093 **tion).** We divide the 100 ms window into three equal sub-
094 windows $[w_0, w_1, w_2]$ of ≈ 33 ms each. For sub-window k
095 ending at t_k^{end} , each event contributes a signed, exponentially
096 decaying weight:

$$097 \quad w_i = p_i \cdot \exp\left(-\frac{t_k^{\text{end}} - t_i}{\tau_{\text{decay}}}\right), \quad \tau_{\text{decay}} = 30 \text{ ms.} \quad (1)$$

098 Events are accumulated into a spatial grid per sub-window
099 and normalised channel-wise as $f(v) = \text{sgn}(v) \log(1 + |v|)$.
100 The resulting 3-channel tensor is compatible with ImageNet-
101 pretrained convolutional kernels.

102 **5-channel signed voxel grid (baseline).** Following Zhu *et*
103 *al.* [18], we divide the window into $B = 5$ temporal bins
104 of 20 ms each and bilinearly interpolate each event across
105 adjacent bins:

$$106 \quad V_k(x, y) = \sum_i p_i \max(0, 1 - |t_i^* - k|) \delta(x - x_i, y - y_i), \quad (2)$$

107 where $t_i^* = (t_i - t_{\min}) / (t_{\max} - t_{\min}) \cdot (B - 1)$ normalises
108 timestamps to $[0, B - 1]$.

109 3.2. Network Architectures

110 **DirectPoseCNN (frame-by-frame).** A ResNet-50 [4]
111 backbone pre-trained on ImageNet1K provides the feature
112 extractor. The first convolutional layer is patched to ac-
113 cept $C = 3$ (or $C = 5$) input channels by averaging the

pretrained RGB weights and replicating them across the
required number of channels, preserving the benefit of ImageNet
initialisation.

Global average pooling yields a 2048-dimensional feature
vector, which feeds into two independent regression heads:

$$\hat{\mathbf{t}} = \mathbf{W}_t^{(2)} \text{ReLU}(\mathbf{W}_t^{(1)} \mathbf{f} + \mathbf{b}_t^{(1)}) + \mathbf{b}_t^{(2)}, \quad (3)$$

$$\hat{\mathbf{q}} = \frac{\mathbf{g}(\mathbf{f})}{\|\mathbf{g}(\mathbf{f})\|_2}, \quad (4)$$

where $\mathbf{f} \in \mathbb{R}^{2048}$, both intermediate layers have 128 units
with ReLU activations and dropout (rate 0.2), and $\mathbf{g}$ maps to
4 units before L2 normalisation enforces the unit-quaternion
constraint.

DomainAdaptivePoseNet (CNN+GRU). To exploit inter-
frame context, we extend DirectPoseCNN with a one-layer
GRU [1] of hidden size 512. A sequence of $T = 10$ con-
secutive event frames is embedded independently by the
shared ResNet-50 backbone (the (B, T, C, H, W) tensor is
reshaped to (BT, C, H, W) for a single forward pass, then
reshaped back to $(B, T, 2048)$). The GRU processes the
sequence and its final hidden state feeds the dual regression
heads.

Domain adaptation is implemented via a gradient-
reversal layer [3] inserted before a binary domain classi-
fier $\text{FC}(512 \rightarrow 64 \rightarrow 2)$. The reversal coefficient λ is
annealed from 0 to 1 following a sigmoid schedule over
training epochs, encouraging the backbone to learn domain-
invariant features.

KeypointHeatmapNet (keypoint-based, future work).
Rather than regressing pose directly, this variant regresses
2D pixel coordinates of $K = 14$ known 3D keypoints on
the Proba-2 satellite, and recovers pose via PnP. Following
the SimpleBaseline architecture [17], we use the ResNet-50
backbone *without* global average pooling (preserving spatial
resolution) and attach a three-stage upsampling decoder:

$$\text{ConvTranspose}(2048 \rightarrow 256, k=4, s=2)^{\times 3} \rightarrow \text{Conv}(256 \rightarrow K, k=1).$$

Each decoder stage is followed by batch normalisation
and ReLU. The output $(B, K, H/4, W/4)$ heatmap is con-
verted to sub-pixel keypoint coordinates via differentiable
soft-argmax with temperature $\beta = 100$. Gaussian target
heatmaps with $\sigma = 2$ px are generated from projected
ground-truth 3D keypoints.

115 3.3. 3D Keypoint Definitions

We define $K = 14$ keypoints on the Proba-2 satellite in its
body frame (metres):

END-TO-END SATELLITE POSE ESTIMATION FROM EVENT DATA

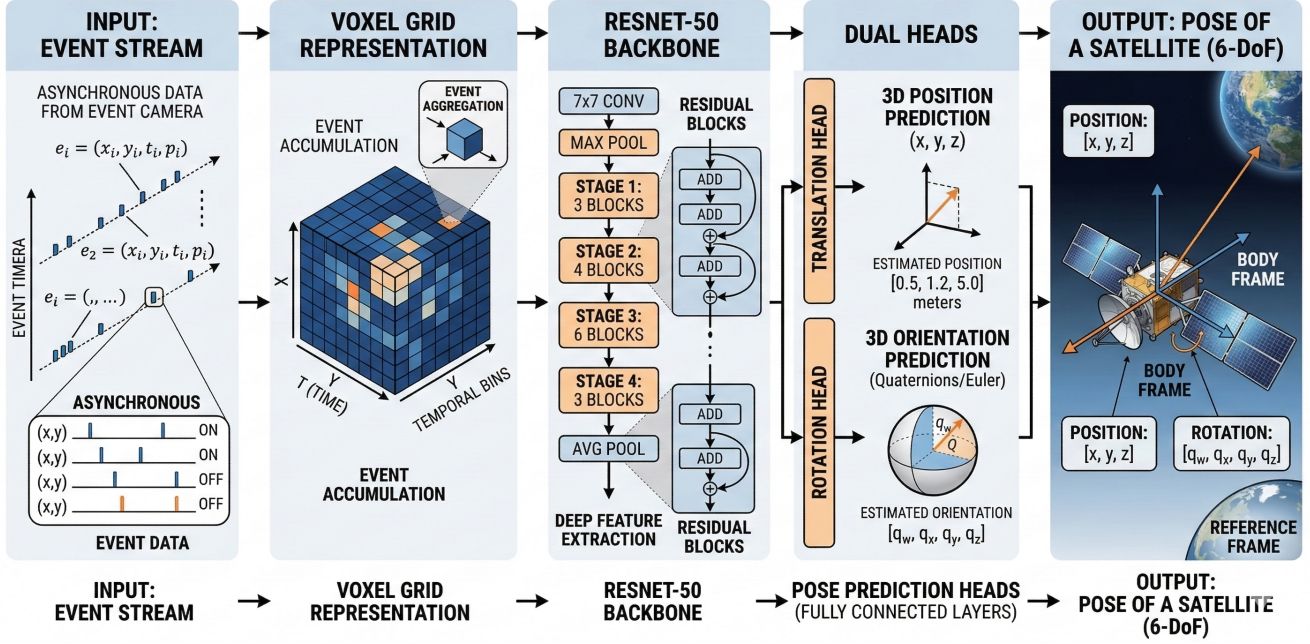


Figure 1. Overview of our end-to-end pipeline. Raw asynchronous events from the event camera are accumulated into a 3-channel exponential-decay voxel grid (or 5-channel signed voxel grid), which is passed through a ResNet-50 backbone. The resulting feature vector feeds dual regression heads producing a 3-D translation vector and a unit quaternion for orientation. Figure generated with AI assistance.

- **8 bounding-box corners:** $[\pm 0.80, \pm 0.30, \pm 0.52]$ — span the full 3D extent for maximum PnP conditioning.
- **6 face centres:** $[\pm 0.80, 0, 0], [0, \pm 0.30, 0], [0, 0, \pm 0.52]$ — ensure at least one visible keypoint under most view-points.

Camera intrinsics: $f_x = f_y = 1258.6$, $c_x = 640$, $c_y = 360$ (no distortion).

3.4. PnP Pose Solver

Given predicted 2D keypoints $\{\hat{\mathbf{u}}_k\}$ and the known 3D coordinates $\{\mathbf{P}_k\}$, we solve for pose using `cv2.solvePnP` with the EPnP initialiser, reprojection threshold $\epsilon = 8$ px, and minimum 4 inliers. The resulting rotation vector is converted to a unit quaternion via `scipy.spatial.transform.Rotation`. If PnP fails (fewer than 4 inlier correspondences), we fall back to the last successfully predicted pose.

3.5. Loss Functions

Composite pose loss. For the direct regression models we minimise:

$$\mathcal{L} = \lambda_t \cdot \|\hat{\mathbf{t}} - \mathbf{t}\|_2^2 + \lambda_r \cdot 2 \arccos(|\hat{\mathbf{q}} \cdot \mathbf{q}|), \quad (5)$$

where the rotation term is the geodesic distance on $\text{SO}(3)$. The absolute value of the quaternion dot product handles the double-cover of $\text{SO}(3)$. We ablate λ_t and λ_r in Sec. 4.4.

Heatmap loss. For KeypointHeatmapNet:

$$\mathcal{L}_{\text{hm}} = \frac{1}{N_{\text{vis}}} \sum_{k=1}^K \mathbf{v}_k \cdot \|\hat{H}_k - H_k^*\|_F^2, \quad (6)$$

where $\mathbf{v}_k \in \{0, 1\}$ is the per-keypoint visibility mask, H_k^* is the Gaussian target, and N_{vis} is the count of visible keypoints.

3.6. Training Strategy

Two-phase curriculum. We train in two phases. *Phase 1* uses a 25% data subset (75 sequences, 20 epochs) for rapid architecture validation; the pass criterion is val. translation $< 10\%$ and rotation < 40 . *Phase 2* fine-tunes on the full 100% dataset (270 training, 30 validation sequences) for up to 100 additional epochs, resuming from the Phase-1 best checkpoint.

Optimiser and schedule. We use AdamW with weight decay 10^{-4} , initial learning rate 5×10^{-6} , gradient clipping at $\|\cdot\|_{\text{max}} = 1.0$, and ReduceLROnPlateau (factor 0.5, patience 5 epochs). Training uses AMP (automatic mixed precision) with gradient scaling. Early stopping triggers after 10 epochs without improvement in validation loss.

Event-camera augmentations. Standard image augmentations (crop, flip) do not transfer meaningfully to event voxel grids. We design augmentations that simulate event-camera failure modes:

1. **RandomIntensityScale:** multiply all voxel values by $s \sim \mathcal{U}(0.8, 1.2)$.
2. **RandomFrameDropout:** zero out an entire temporal channel with probability 0.1 (simulates event-rate drops).
3. **SaltPepperNoise:** randomly set 1–2% of spatial locations to ± 1 (hot/dead pixels), applied with probability 0.7.
4. **RandomErasing:** zero a random rectangle covering 10–30% of the frame area, with probability 0.5 (simulates packet loss).

No augmentations are applied at validation or inference time.

4. Experiments

4.1. Dataset and Evaluation

The SPADES dataset [12] contains 300 synthetic training sequences and 31 real test sequences captured with a 1280×720 Prophesee event camera observing a Proba-2 spacecraft mockup in a controlled environment. Approximately 179 000 training frames are available after preprocessing at the 10 Hz pose rate. We hold out 10% of training sequences (30 sequences) for validation.

The competition metric is the composite pose error:

$$E = E_t + E_r, \quad E_t = \frac{\|\hat{\mathbf{t}} - \mathbf{t}\|}{\|\mathbf{t}\|}, \quad E_r = 2 \arccos(|\hat{\mathbf{q}} \cdot \mathbf{q}|) \text{ [rad]}, \quad (7)$$

evaluated over all 31 test sequences.

Timestamp scaling. Synthetic H5 files store timestamps in 100 μs units, while real test recordings use 1 μs units. This 100× mismatch compresses the temporal distribution of events within voxel bins by a factor of 100 at test time, constituting a severe domain shift. All inference runs use `--test-timestamp-scale 1.0` to apply the real-data unit convention.

4.2. Implementation Details

Preprocessing converts raw H5 event files into voxel grids stored as HDF5 at 256×448 (Phase 1) and 1280×720 (Phase 2) resolution, requiring approximately 1.4 MB and 11 MB per frame respectively. Training runs on a single NVIDIA A100-SXM4-80GB GPU. At batch size 16 and full 100% resolution, one epoch over 179 K frames takes approximately 2.5 hours.

4.3. Main Results

Tab. 1 summarises all submitted results alongside the SPADES dataset paper baselines [12].

Table 1. Test-set results on SPADES. E_t : relative translation error; E_r : geodesic rotation error (rad); E : composite score. Lower is better. SPADES baselines are the Direct and Hybrid approaches reported in [12] on the same real test sequences.

Method	E_t	E_r (rad)	E
SPADES Direct [12]	0.0513	1.416	1.47
SPADES Hybrid [12]	0.0334	1.378	1.41
CNN+GRU, $T=10$ (ours)	0.6722	1.6113	2.2835
DirectPoseCNN, $\lambda_t=2.0$, $\lambda_r=1.0$ (ours)	0.4105	1.6526	2.063

Frame-by-frame beats temporal modelling. The CNN+GRU model scored 2.2835 ($E_t = 0.6722$, $E_r = 1.6113$ rad) vs. 2.063 ($E_t = 0.4105$, $E_r = 1.6526$ rad) for DirectPoseCNN. Translation error is 64% higher under the temporal model, while rotation error is marginally better (1.6113 vs. 1.6526 rad), suggesting the GRU captures orientation context but degrades translation predictions. At a 10 Hz sampling rate with 100 ms voxel windows, consecutive frames share approximately 90% of their event window, providing minimal new temporal information while increasing optimisation difficulty.

4.4. Loss Weight Ablation

We ablate the loss weights (λ_t , λ_r) while holding all other hyperparameters fixed, resuming from the same Phase-1 checkpoint to ensure a fair comparison.

Table 2. Validation metrics for different loss weight combinations. All runs resume from the same epoch-29 checkpoint.

λ_t	λ_r	Val E_t (%)	Val E_r (°)	Note
2.0	1.0	14.64	92.46	Submitted (best)
1.5	2.0	15.01	91.58	Best val rotation
0.0	1.0	≈15.2	≈92.5	Frozen trans. head

Increasing λ_r relative to λ_t yields modest improvements in validation rotation error (0.88° at best) but at the cost of elevated translation error (+0.37 pp). Freezing the translation head while training the rotation head and backbone causes the backbone features to shift, degrading translation despite the frozen FC head. This confirms that the shared backbone tightly couples both tasks.

The best rotation result (91.58) was achieved at epoch 37 with $\lambda_t = 1.5$, $\lambda_r = 2.0$; however, the improvement was not submitted due to time constraints.

4.5. Domain Gap Analysis

Tab. 3 and Fig. 2 quantify the validation-to-test domain gap.

Translation error is amplified by 2.77× from validation to test, while rotation error increases by only 2.23°. The large

Table 3. Synthetic-to-real domain gap for our best DirectPoseCNN.

Metric	Val (synthetic)	Test (real)
Translation error	14.64%	41.05%
Rotation error	92.46°	94.69°
Amplification	$\times 2.77$	$+2.23^\circ$

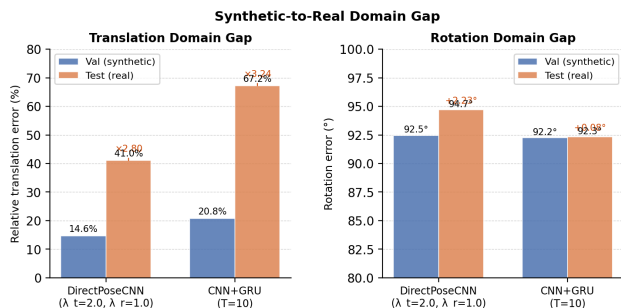


Figure 2. Synthetic-to-real domain gap for both submitted models. Translation error amplifies by $2.77\times$ (DirectPoseCNN) and $3.24\times$ (CNN+GRU) from validation to test, while rotation error changes by less than 0.5, indicating the domain shift primarily affects translation.

translation amplification likely stems from the timestamp-unit mismatch ($100\times$), which alters the density of events per voxel bin. Because rotation depends primarily on the *spatial pattern* of events and translation depends on absolute scale, the former is more robust to temporal rescaling.

4.6. Leaderboard Context

Tab. 4 shows the final public leaderboard.

Table 4. Final SPARKS 2026 leaderboard (top 5 and our entry).

Rank	Team	E_t	E_r (rad)	E
1	National Key Lab	0.0146	0.0355	0.0501
2	NSSC_UCAS	0.0230	0.0331	0.0561
3	haoranhuan	0.1464	0.5562	0.7026
4	pruthj	0.1335	0.9101	1.0436
5	csu_nuaa_pang	0.3340	0.7724	1.1064
12	santhoshsrs (ours)	0.4105	1.6526	2.063

The top two teams achieve near-perfect translation ($E_t \approx 0.02$), suggesting they employ model-based methods or exploit explicit 3D spacecraft geometry rather than relying on learned translation from data alone.

5. Discussion

Geodesic loss is necessary. Preliminary experiments with ℓ_2 quaternion loss failed to converge reliably (rotation errors > 150). Geodesic distance in Eq. (5) provides well-defined

gradients across the entire $SO(3)$ manifold and handles the sign ambiguity of quaternion representation.

Timestamp normalisation is critical for domain transfer.

The $100\times$ timestamp-unit mismatch between synthetic and real data is the dominant source of domain shift. Normalising timestamps within each event window to a fixed range $[0, T_{\text{window}}]$ before constructing voxel grids would make the representation invariant to this shift and is a straightforward improvement for future work.

Temporal modelling does not help at 10 Hz.

With 100 ms voxel windows and 10 Hz pose sampling, adjacent frames are highly correlated. Temporal modelling with a GRU introduces additional complexity without providing new information. A higher pose sampling rate or shorter event windows might benefit from temporal aggregation.

GlobalAvgPool destroys spatial information for key-points.

Our initial keypoint regressor, which applied GlobalAvgPool before the regression head, was fundamentally limited to ≈ 160 px keypoint error because all spatial information was discarded. Switching to a heatmap decoder (removing the pooling operation) is the correct architectural choice; preliminary experiments showed expected error of 20–40 px. Future work should complete the heatmap training and evaluate the full keypoint+PnP pipeline.

Loss weight coupling.

Because translation and rotation heads share the ResNet-50 backbone, tuning λ_t and λ_r creates implicit trade-offs: improving rotation can hurt translation and vice versa. Disentangling these tasks, for instance with separate backbone branches or task-specific normalisation layers, could provide more independent control.

6. Conclusion

We presented an event-camera spacecraft pose estimation pipeline evaluated on the SPADES dataset as part of the SPARKS Challenge at CVPR 2026. Our best result, a composite pose error of **2.063**, is achieved by a ResNet-50 direct regression model trained with geodesic rotation loss on 3-channel exponential-decay event voxel grids. We found that: (i) single-frame prediction outperforms temporal modelling at 10 Hz; (ii) synthetic-to-real translation error is amplified $\approx 2.8\times$ at test time due to a $100\times$ timestamp-unit mismatch; and (iii) loss weight tuning for the rotation head is constrained by the shared backbone. Future work will focus on timestamp-invariant voxel construction, a completed heatmap keypoint regressor with PnP, and explicit domain normalisation to reduce the synthetic-to-real gap.

References

- [1] Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using rnn encoder-decoder for statistical machine translation. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1724–1734, 2014. 2
- [2] Guillermo Gallego, Tobi Delbrück, Garrick Orchard, Chiara Bartolozzi, Brian Taba, Andrea Censi, Stefan Leutenegger, Andrew J Davison, Jörg Conradt, Kostas Daniilidis, et al. Event-based vision: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(1):154–180, 2022. 1
- [3] Yaroslav Ganin, Evgeniya Ustun, Muhammad Ghifary, Victor Lempitsky, et al. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59):1–35, 2016. 2
- [4] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016. 2
- [5] Yinlin Hu, Joachim Hugonot, Pascal Fua, and Mathieu Salzmann. Segmentation-driven 6d object pose estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3385–3394, 2019. 1
- [6] Alex Kendall, Matthew Grimes, and Roberto Cipolla. Posenet: A convolutional network for real-time 6-dof camera relocalization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2938–2946, 2015. 1
- [7] Xavier Lagorce, Garrick Orchard, Francesco Galluppi, Bertram E Shi, and Ryad B Benosman. Hots: A hierarchy of event-based time-surfaces for pattern recognition. In *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pages 1346–1359, 2017. 1
- [8] Siddharth Mahendran, Haider Ali, and René Vidal. 3d pose regression using convolutional neural networks. *arXiv preprint arXiv:1708.05628*, 2017. 2
- [9] Kiru Park, Timothy Patten, and Markus Vincze. Pix2pose: Pixel-wise coordinate regression of objects for 6d pose estimation. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 7668–7677, 2019. 1
- [10] Tae Ha Park, Marcus Mörtens, Gurvan Lecuyer, Dario Izzo, and Simone D’Amico. Speed+: Next-generation dataset for spacecraft pose estimation across domain gap. *arXiv preprint arXiv:2110.03101*, 2022. 1
- [11] Valentin Peretroukhin, Matthew Giamou, David M Rosen, W Nicholas Greene-Diniz, Jonathan Kelly, and John J Leonard. A smooth representation of belief over $so(3)$ for deep rotation learning with uncertainty. In *Robotics: Science and Systems*, 2020. 2
- [12] Arunkumar Rathinam, Haytam Qadadri, and Djamila Aouada. SPADES: A realistic spacecraft pose estimation dataset using event sensing. *arXiv preprint arXiv:2311.05310*, 2023. 2, 4
- [13] Henri Rebecq, Timo Horstschäfer, Guillermo Gallego, and Davide Scaramuzza. Real-time visual-inertial odometry for event cameras using keyframe-based nonlinear optimization. In *British Machine Vision Conference (BMVC)*, 2017. 1
- [14] Henri Rebecq, Rene Ranftl, Vladlen Koltun, and Davide Scaramuzza. Events-to-video: Bringing modern computer vision to event cameras. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3857–3866, 2019. 1
- [15] Sumant Sharma and Simone D’Amico. Pose estimation for non-cooperative spacecraft rendezvous using convolutional neural networks. In *2019 AAS/AIAA Astrodynamics Specialist Conference*, 2019. 1
- [16] SPARKS Challenge Organizers. SPARKS challenge: Spacecraft pose estimation from event cameras, 2026. Workshop at CVPR 2026. 1
- [17] Bin Xiao, Haiping Wu, and Yichen Wei. Simple baselines for human pose estimation and tracking. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 466–481, 2018. 2
- [18] Alex Zihao Zhu, Liangzhe Yuan, Kenneth Chaney, and Kostas Daniilidis. Unsupervised event-based learning of optical flow, depth, and egomotion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 989–997, 2019. 1, 2